Smart Medi Box - Wiring Guide

Professional Edition

System Overview

The Smart Medi Box runs from a 12V supply split into two rails for power isolation.

12V Rail (Direct)

- Solenoid Lock
- Optional Cooling System

5V-6V Rail (via buck converter)

- Servo Motor

5V Rail (via regulator)

- Arduino Leonardo
- SIM800L EVB
- Sensors (DHT22, DS18B20, RTC)
- LCD Display
- RFID Module (via 3.3V pin)

Power System Design

Input Supply

`12V / 5A DC Power Supply (recommended)`

Voltage Regulation

Recommended: Buck Converter (LM2596)
Alternative: LM7805 (with heatsink)

Capacitors (Essential)

- Input: 100uF electrolytic
- Output: 10uF electrolytic + 100nF ceramic
- SIM800L: 1000uF near VCC

> Linear regulators generate heat: Power Loss = (12V - 5V) x Current. Use a buck converter for efficiency.

Critical rule: All components must share a common ground (GND).

Microcontroller (Arduino Leonardo)

Power

- VCC -> 5V
- GND -> Common Ground

Pin Configuration

Digital Pins

Pin	Function
0	SIM800L RX
1	SIM800L TX
2	Door Switch
3	Cooling (MOSFET Gate)
4	Stepper STEP
5	Solenoid (MOSFET Gate)
6	RFID RST
7	Buzzer
8	LCD Reset
9	RFID CS
10	LCD RS
11	LCD RW
12	Stepper DIR

Pin	Function
13	LCD E

Analog Pins

Pin	Function
A0	DHT22 Data
A1	DS18B20 Data
A2	Servo Signal (MG995)

Hardware Compatibility Check

- SIM800L EVB: Requires stable 5V with high peak current. Keep 1000uF cap close to VCC/GND.
- RC522 RFID: Must use 3.3V logic. Do not connect VCC to 5V. Keep SPI lines short.
- ST7920 LCD: Uses ST7920 serial (not standard SPI). Verify PSB to GND for serial mode.
- DHT22 + DS18B20: Use 4.7K pull-up on data. Keep data lines away from motor wiring.
- Servo MG995: Use 5V-6V only. Do not power from 12V rail.
- Solenoid: Must use flyback diode across coil. MOSFET required.
- Shared GND: All modules must share a common ground reference.
- Door switch on D2 conflicts with I2C SDA. If RTC is used, move the door switch to a free pin.

Power Wiring Diagram (ASCII)

```
12V DC IN
   |
   +--------------------> Servo (5V-6V supply via buck)
   |
   +--------------------> Solenoid (12V) -> MOSFET -> GND
   |
   +--> Buck Converter (LM2596) -> 5V RAIL
           |
           +--> Arduino Leonardo 5V
           +--> SIM800L EVB 5V (1000uF cap near VCC/GND)
           +--> LCD 5V
```

```
            +--> Sensors 5V
            +--> Stepper Driver (5V logic + separate motor supply 6V-12V)
            +--> RC522 3.3V (from Arduino 3.3V pin)

ALL GROUNDS TIED TO COMMON GND
```

Arduino Leonardo Full Pinout (Reference)

Digital Pins

Pin	Function	Notes
0	RX	UART RX (Serial1)
1	TX	UART TX (Serial1)
2	SDA	I2C SDA
3	SCL	I2C SCL
4	GPIO	Digital I/O
5	PWM	Digital I/O, PWM
6	PWM	Digital I/O, PWM
7	GPIO	Digital I/O
8	GPIO	Digital I/O
9	PWM	Digital I/O, PWM
10	PWM	Digital I/O, PWM
11	PWM	Digital I/O, PWM
12	GPIO	Digital I/O
13	LED	Digital I/O, onboard LED

Analog Pins

Pin	Function	Notes
A0	Analog In	ADC0
A1	Analog In	ADC1
A2	Analog In	ADC2
A3	Analog In	ADC3
A4	Analog In	ADC4
A5	Analog In	ADC5

Power Pins

Pin	Function
VIN	External input (7V-12V)
5V	Regulated 5V output
3.3V	3.3V output (for RC522)
GND	Ground
RESET	Reset input

SPI Pins

- **IMPORTANT (Arduino Leonardo):** Use ICSP header for SPI (MOSI, MISO, SCK).
- D11-D13 are not used for SPI on Leonardo. Use ICSP for RC522.

USB Serial

- USB serial is separate from pins 0/1 on Leonardo.

SIM800L EVB (GSM Module)

Connections (4 wires only)

Pin	Connection
VCC	5V
GND	GND
TX	Arduino Pin 0 (RX)
RX	Arduino Pin 1 (TX)

Note: TX/RX must be crossed (TX -> RX, RX -> TX).

Requirements

- SIM card (correct orientation)
- Antenna connected
- 1000uF capacitor for stability

Important: SIM800L draws up to 2A peak current. Ensure:
- Thick power wires
- Short connection to power source
- 1000uF-2200uF capacitor near module
- Do NOT power from Arduino 5V pin

LCD Display (12864B V2.3 - ST7920)

LCD Type & Mode

- **Model:** ST7920 128×64
- **Mode:** Serial (SPI-like mode) when PSB = GND
- ⚠️ **IMPORTANT:** PSB must be GND for serial mode. If PSB = 5V, it switches to parallel mode and U8g2 wiring/constructor must change.

Connections

LCD Pin	Function	Arduino Pin
PSB	Mode Select	GND (Serial mode)
VCC	Power	5V

LCD Pin	Function	Arduino Pin
GND	Ground	GND
RS	Register Select	Pin 10
RW	Read/Write	Pin 11
E	Enable	Pin 13
RST	Reset	Pin 8

Library: U8g2

U8g2 Constructor (Working - Serial/SPI Mode)

```
#include <U8g2lib.h>
U8G2_ST7920_128X64_F_SW_SPI u8g2(U8G2_R0, 13, 11, 10, 8);
// Parameters: (rotation, clock=Pin13(E), data=Pin11(RW), cs=Pin10(RS), reset=Pin8(RST))
```

Uses ST7920 serial protocol (not standard SPI). PSB = GND → Serial mode (this config). PSB = 5V → Parallel mode (requires different constructor & wiring).

Backlight

- 🔆 **BLA** (Backlight Anode, +) → 5V
- ⚫ **BLK** (Backlight Cathode, −) → GND

Sensors

DHT22 (Temperature & Humidity)

Required Libraries: DHT sensor library, OneWire, DallasTemperature

Wiring:

- VCC → 5V
- GND → GND
- DATA → A0
- **4.7K pull-up resistor required** between DATA and 5V (or 3.3V)

DS18B20 (Temperature)

Wiring:

- VCC → 5V
- GND → GND
- DATA → A1
- **4.7K pull-up resistor required** between DATA and 5V (or 3.3V)

RTC DS3231 (I2C)

Required Library: RTClib by Adafruit

DS3231 Wiring on Leonardo:

- Use the **SDA/SCL labeled pins** (near AREF), NOT D2/D3
- DS3231 SDA → SDA header pin
- DS3231 SCL → SCL header pin
- VCC → 5V
- GND → GND

> ⚠️ Important: On Leonardo, use the dedicated SDA/SCL header pins (near AREF), not the D2/D3 digital pins. This avoids conflicts with the door switch on D2.

RTC Time Set (Compile Time)

```
#include <Wire.h>
#include <RTClib.h>

RTC_DS3231 rtc;

void setup() {
  Serial.begin(9600);
  delay(1000);

  if (!rtc.begin()) {
    Serial.println("Couldn't find DS3231 (check SDA/SCL pins).");
    while (1) {}
  }

  Serial.println("RTC will be set to COMPILE TIME after countdown.");
  for (int i = 10; i >= 1; i--) {
    Serial.print("Setting in ");
    Serial.println(i);
    delay(1000);
  }
```

```
  rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
  Serial.println("Done. RTC set.");
}

void loop() {}
```

RFID Module (RC522)

Connections (ICSP Header)

RC522 Pin → Arduino Pin/Header:

- RC522 MOSI → ICSP MOSI
- RC522 MISO → ICSP MISO
- RC522 SCK → ICSP SCK
- RC522 VCC → **3.3V** (NOT 5V)
- RC522 GND → GND
- RC522 SDA/SS (CS) → Pin 9
- RC522 RST → Pin 6

> ⚠️ **CRITICAL NOTES:**
> ✓ RC522 operates at 3.3V logic level. Do NOT connect VCC to 5V.
> ✓ Use ICSP header for SPI (not digital pins D11-D13 on Leonardo).
> ✓ ICSP pinout diagram is attached in the build pack.
> ✓ Consider using a logic-level converter or resistor dividers for MOSI, SCK, and CS lines if power supply is not stable.

Motors and Actuators

Stepper Motor (EasyDriver HW-135)

- STEP -> Pin 4
- DIR -> Pin 12
- VCC -> 5V (logic)
- M+ -> 6V-12V (motor supply)
- GND -> GND

Servo Motor (MG995)

- Signal → A2 (digital pin 20)
- VCC → 5V–6V (NOT 12V)
- GND → GND

> On Leonardo, analog pins can be used as digital I/O. Using `s.attach(A2)` is valid (A2 is also known as digital pin 20). Use external 5V supply if needed.

Solenoid Lock (12V)

- Positive -> 12V
- Negative -> MOSFET Drain

MOSFET (N-Channel)

- Gate -> Pin 5 (via 1K resistor)
- Drain -> Solenoid
- Source -> GND

Protection

- Diode (1N4007) across solenoid

Buzzer (TMB12A05 + BC337 Driver)

Buzzer Type: Active electromagnetic buzzer

Wiring:

- Arduino D7 → 1K resistor → BC337 BASE
- BC337 EMITTER → GND
- BC337 COLLECTOR → Buzzer (-)
- Buzzer (+) → +5V

Protection Diode (Recommended):

- Use 1N4148 or 1N4007 across the buzzer coil
- Diode cathode (stripe) → +5V (buzzer +)
- Diode anode → Buzzer (-) / BC337 collector

Common Ground: Arduino GND must be connected to the buzzer supply GND

Operation: This is an ACTIVE buzzer—it makes sound with DC voltage

- Use `digitalWrite(D7, HIGH)` to turn ON
- Use `digitalWrite(D7, LOW)` to turn OFF

Cooling System (Pin D3) – MOSFET Low-Side Switch

Supported Loads:

- DC fan (2-wire)
- Peltier / TEC module (thermoelectric cooler)
- Any DC load within MOSFET + power supply limits

Wiring (LOW-SIDE MOSFET SWITCH):

- Arduino D3 → 1K resistor → MOSFET GATE
- MOSFET SOURCE → GND (power supply negative)
- MOSFET DRAIN → Load (-) (fan- or TEC-)
- Load (+) → +V supply (12V typical for fan/TEC)

CRITICAL REQUIREMENTS:

- **1) Common Ground Required:** Arduino GND must be connected to the load power supply GND
- **2) Gate Pulldown (Recommended):** Add 100K (or 10K) resistor from MOSFET GATE to GND to prevent floating gate
- **3) Flyback Diode:** REQUIRED for inductive loads (DC fan, motor, relay). NOT used for TEC/Peltier.
- **4) Peltier (TEC) WARNING:** TEC modules draw HIGH current (3A–10A+). Use a separate PSU and logic-level MOSFET. Consider a fuse.
- **5) PWM Control:** D3 supports PWM. For TEC, PWM can reduce efficiency; consider a thermostat/PID approach for stable temperature.

Door Switch (Pin D2)

Pin Configuration: Pin 2 with INPUT_PULLUP

```
const int DOOR_PIN = 2;

void setup() {
  Serial.begin(9600);
```

```
  pinMode(DOOR_PIN, INPUT_PULLUP);
}

void loop() {
  int state = digitalRead(DOOR_PIN);
  if (state == LOW) {
    Serial.println("Door CLOSED (switch closed to GND)");
  } else {
    Serial.println("Door OPEN (switch open)");
  }
  delay(100);
}
```

Power Budget

5V Rail

Approx: 350mA typical | Peak: ~600mA

12V Rail

Servo + Solenoid: up to ~1.8A

SIM800L

Peak bursts: ~2A

Power supply: 12V / 5A recommended (SIM800L + servo peaks)

Protection (Recommended)

- Add inline fuse (2A-5A) on 12V input
- Add reverse polarity protection diode

Key Design Rules

- All grounds must be connected together
- Use 3.3V for RFID (RC522)
- Servo must use 5V-6V only (not 12V)

- Add capacitors for stability
- Avoid pin conflicts
- Use short wires for communication lines
- Linear regulators generate excessive heat (use buck converter)

First Power-On Procedure

1. Verify 5V and 12V using multimeter
2. Power Arduino only -> test code
3. Connect SIM800L -> test AT command
4. Add sensors one by one
5. Test actuators last

Troubleshooting Essentials

No GSM Response

- Check TX/RX wiring
- Check power stability
- Verify baud rate (115200)

Sensor Failure

- Check pull-up resistors
- Verify pin connections

Motor Issues

- Check driver wiring
- Verify control signals

System Resets

- Add bulk capacitor (1000uF)
- Improve grounding

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